

SWAPCELL · REQUIREMENTS

SwapCell requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/swapcell		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3. Record Amish's 2026-09-25 decisions (SWC-DDR-001); add R13 to R15 for interface v0.3 (wake, charge-discharge mode, latch class V1) and R16 budget scope; status from SWC-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2 (electrical, mechanical, interface, protection)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SwapCell requirements

These are the requirements for the pack, the dock and the SwapCell interface v0.3 between them. Amish accepted the TRL 2 recommendations on 2026-09-25 (SWC-DDR-001), including three interface additions, now R13 to R15. Requirements R6, R10 and R12 to R15 define the open interface that other portfolio designs depend on, so changes to them need Amish's approval. The status column comes from SWC-CAL-001; no requirement is shown to be not met, and three are at risk.

ID	Requirement	Target	Verification	Status (SWC-CAL-001)
R1	Nominal voltage in the 48 V light-vehicle class	46.8 V nominal; operating window 39.0 to 54.6 V (3.0 to 4.2 V per cell)	Cell configuration calculation; datasheet	Met
R2	Usable capacity and energy	10 Ah or more and 450 Wh or more at 0.2C, 25 °C	Calculation from cell datasheet; later capacity test. Applies to the standard 4.2 V charge; fleet mode (4.1 V) gives about 419 Wh by design	At risk: 9.7 Ah at the cell datasheet minimum
R3	Discharge current, continuous and peak	20 A continuous for a full discharge with cells below 60 °C from 25 °C ambient; 35 A peak for 10 s	Thermal and cell-current calculation; later load test	At risk: 51 °C in open air, 63 °C enclosed
R4	Charge time at the standard dock	3 h or less from empty to full at 5 A; 80 % in 2 h or less	CC-CV charge calculation	Met
R5	Mass for one-hand carry and swap	3.5 kg or less (7.7 lb) including handle and connector	Mass budget; later weighing	Met: 2.85 kg
R6	Mechanical envelope (interface)	Body 340 x 90 x 80 mm, +0 / -1.5 mm; overall length with handle and plug 400 mm or less	Massing model, then drawing sheet	Met: 393 mm overall
R7	Swap time and effort	Pack out of a dock or vehicle and a charged pack seated and latched in 10 s or less, one hand, no tools; insertion force 50 N or less	Design review; later timed trial	Not verifiable at TRL 3
R8	Ingress protection	Pack IP65, including the unmated connector face; dock IP54 for sheltered outdoor mounting	Design review of seals and contacts	Not verifiable at TRL 3
R9	Cycle life and state-of-health tracking	500 cycles or more to 80 % of rated capacity at 0.5C charge and discharge, 25 °C; SoH estimate within ± 5 %	Cell datasheet review; later cycling data. The 4.1 V fleet charge mode (decided) is the main mitigation	At risk
R10	Blind-mate connector (interface)	5,000 mating cycles or more; self-aligns over ± 3 mm lateral and $\pm 2^\circ$ angular misalignment; 40 A continuous per power contact; contact sequence PACK- first, INTERLOCK last	Connector datasheet; design review	Not verifiable at TRL 3
R11	Protections	Cell overvoltage and undervoltage, charge and discharge overcurrent, short-circuit trip within 500 μ s, over-temperature, charge lockout below 0 °C, cell-level fusing, pre-charge, output dead when unmated	BMS specification review; later fault injection	Not verifiable at TRL 3
R12	Data interface and logging (interface)	CAN 2.0B at 250 kbit/s with a published message set; pack stores 2,000 or more cycle records; any conforming dock reads and exports them as CSV	Message set review, bus-load calculation; later bench decode	Met on paper

ID	Requirement	Target	Verification	Status (SWC-CAL-001)
		without a cloud account; bit layout as published in SWC-PRC-001 v0.3		
R13	Wake for hosts without CAN or a wake supply (interface v0.3 item W)	Pack wakes from sleep when a receiver closes INTERLOCK through a 10 kΩ coding resistor, drawing no supply from the host; also wakes on WAKE (5 to 15 V) or its manual button; rejects a shorted or open loop; sleep drain 1 % of capacity per month or less; legacy discharge-only mode at 15 A for hosts without CAN	Calculation; later bench check	Met on paper
R14	Charge-while-discharging mode (interface v0.3 item C)	Dock, station and vehicle hosts may request mode 4 (charge-discharge); pack allows net current between the allowed charge and allowed discharge limits without interrupting its output, and on a charge-side fault stops charge only and keeps discharge	Message set review; calculation; later bench check	Met on paper
R15	Latch vibration rating for vehicles (interface v0.3 item V)	Class V1 vehicle receivers and the pack latch: no latch release and no power-contact interruption of 1 ms or longer under the UN 38.3 test T3 sine profile (7 to 200 Hz, peak 8 g) in three axes and 25 g, 11 ms half-sine shocks; latch proof load 1.72 kN along the insertion axis; receiver preload 330 N or more with 50 N or less hand force. Class D (gravity docks) needs no rating	Calculation; later vibration test (TRL 4, on hold)	Not verifiable at TRL 3
R16	Prototype budget scope	Parts for one pack and one wall dock cost \$700 or less (<code>budget_usd</code> in <code>project.yaml</code>). A SwapCell pack is priced once, here, and excluded from each dependent kit budget	Priced BOM	Met: \$559

■ Assumptions

- A 48 V class pack is 13 lithium-ion cells in series. The 3.0 V per cell lower limit is conservative for most NMC and NCA cells; the BMS enforces it regardless of the vehicle controller cutoff.
- 20 A continuous (about 940 W) covers a 750 W cargo-trike motor with margin. E-bikes at 250 to 500 W draw 5 to 11 A.
- A fleet vehicle swaps once or twice a day. Over five years that is about 3,650 mating cycles, so R10 at 5,000 cycles has about 37 % margin (SWC-CAL-001). Dock-side contacts see more cycles than pack-side ones and should be replaceable.
- R5 and R7 assume an adult lifting from a wall dock at 0.8 to 1.2 m height. A lower mounting height would need a review of R5.
- R9 depends on the cell chosen. High-energy 21700 cells often quote fewer cycles than high-power cells; see SWC-PRC-001.
- R13 to R15 were approved by Amish on 2026-09-25 as interface v0.3 additions. The specific values (10 kΩ coding, 100 μA sleep target, 25 g shock, 330 N preload) are engineering proposals inside those approved additions and await his confirmation (SWC-DDR-001 item 19).

- R15 uses the UN 38.3 test T3 vibration profile because every pack must pass it for transport anyway; the 25 g shock is a proposed curb-strike level for rigid cargo vehicles.
- R16: the \$700 budget covers paper design at TRL 3 and the parts for one pack and one dock. A second pack for a real swap demonstration (about \$414 more) needs a build budget decision, which is open and only relevant once TRL 4 is lifted.

Safety: The pack is a 468 Wh lithium-ion battery at up to 54.6 V DC that can deliver about 500 A into a short. R11, R13 (reject a shorted INTERLOCK loop) and R14 (never charge below 0 °C, even in charge-discharge mode) are safety requirements, not conveniences.